



Requirements *for* LIFE

THE ESSENTIALS OF LIFE are very simple. All organisms need energy, water, shelter, and space *to grow*. The majority of life forms also need oxygen, nutrients, and a comfortable range of temperatures.

ENERGY

Without energy all living things would not be able to grow or function. The primary source of energy on Earth is light from the Sun. Animals cannot use it directly but plants and a few other organisms, such as green algae, are able to capture it and use it to make food. Plants are eaten by animals and these animals become food for carnivores (animals that only eat meat) and omnivores (animals that eat both plants and meat).

WATER

All living things need water. Cells are mostly made of water, and water plays a key role in moving vital substances in and out of the cell. Some organisms can survive with very little water. Desert-dwelling plants and animals, such as cacti and camels, have mechanisms that help them cope with drought and capture as much water as possible when it is available. By contrast, fish and other aquatic animals spend their whole life in water.

SHELTER

At some point in their lives, most animals search out somewhere for shelter (see pp.60–61). They do this to avoid predators or bad weather, to go to sleep, or to give birth to their young in safety. Without shelter, they could die from exposure or become dinner for another animal. Plants aren't so lucky—because they can't move they have to find different ways to tough out the weather and fend off animals that want to eat them.

